NAME: ALLAN NJUGUNA NGUGI
EMAIL: ngugiallan05@gmail.com
ADMISSION NUMBER: CS-SA12-26007

Table of Contents

Introduction 3

Nessus Skills Assessment 3

OpenVAS Skills Assessment 6

Screenshot Showing Module Completion 12

Conclusion 12

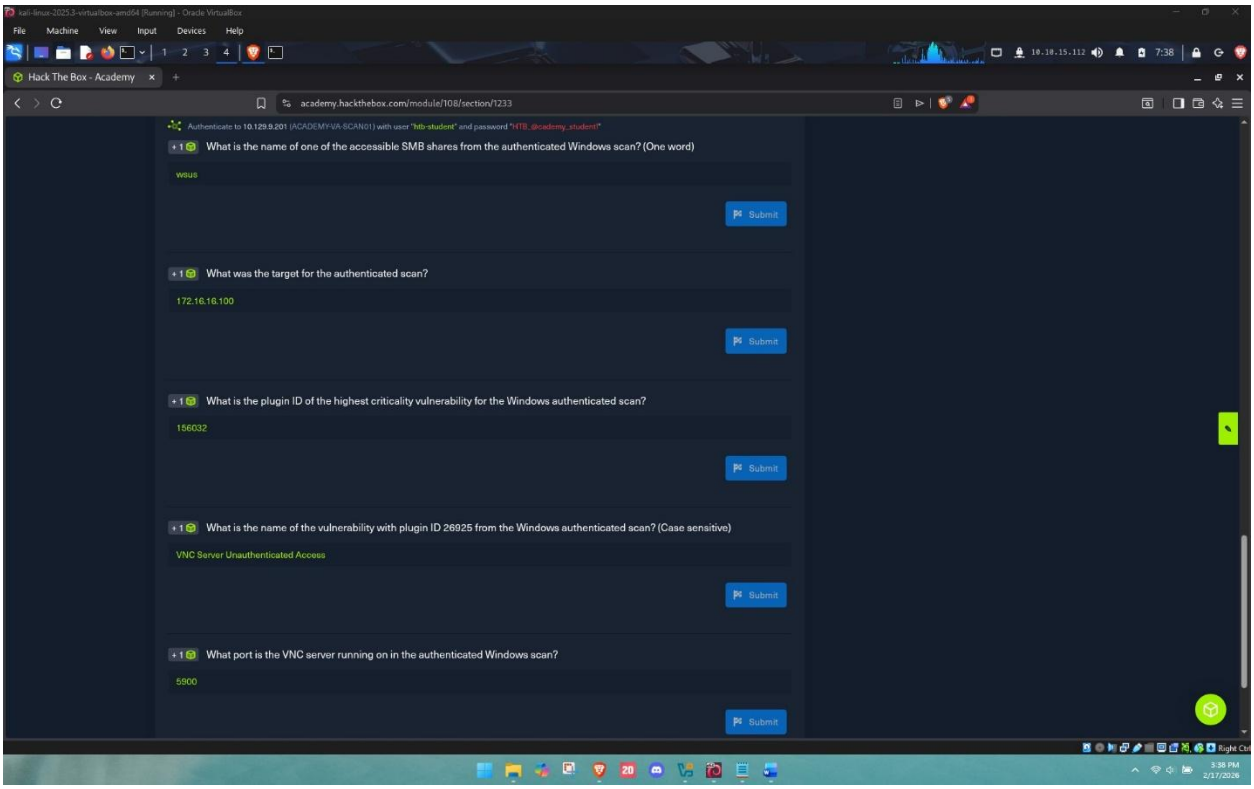
Link Showing Module Completion 12

Introduction

A vulnerability assessment exists as a method for discovering and assessing security weaknesses across computer systems and networks and application software before attackers gain access to those weaknesses. The process includes system scanning and risk assessment and security improvement recommendations which together form the complete assessment. The process protects organizations from cyberattacks while safeguarding their sensitive information because organizations use Nessus and OpenVAS as assessment tools.

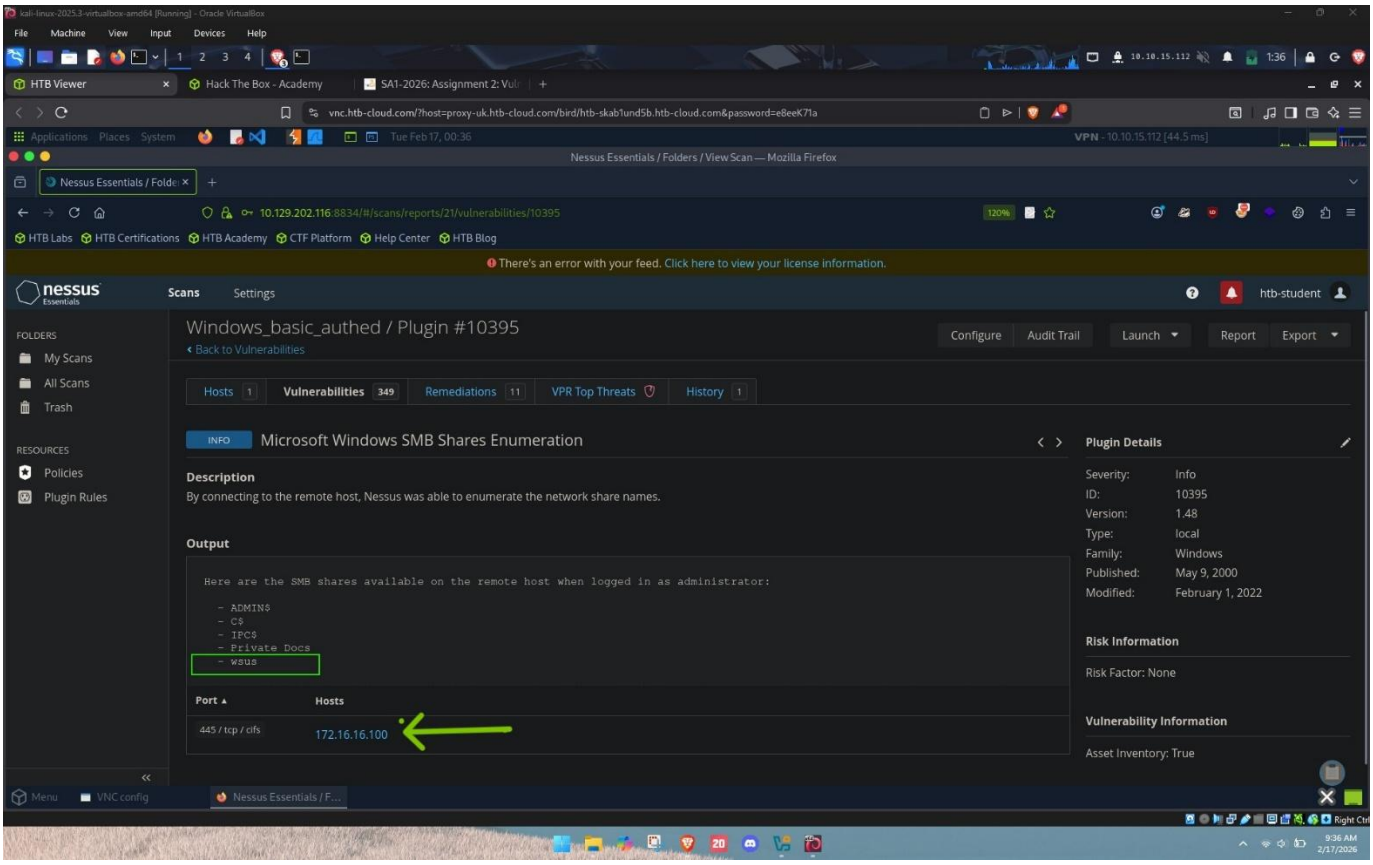
Nessus Skills Assessment

Nessus serves as a vulnerability scanning tool which Tenable developed to detect security flaws in computer systems, network infrastructure, and software applications. The system performs scans to detect known security vulnerabilities and system configuration errors and uninstalled software updates, generating reports that assist organizations in addressing their security threats.

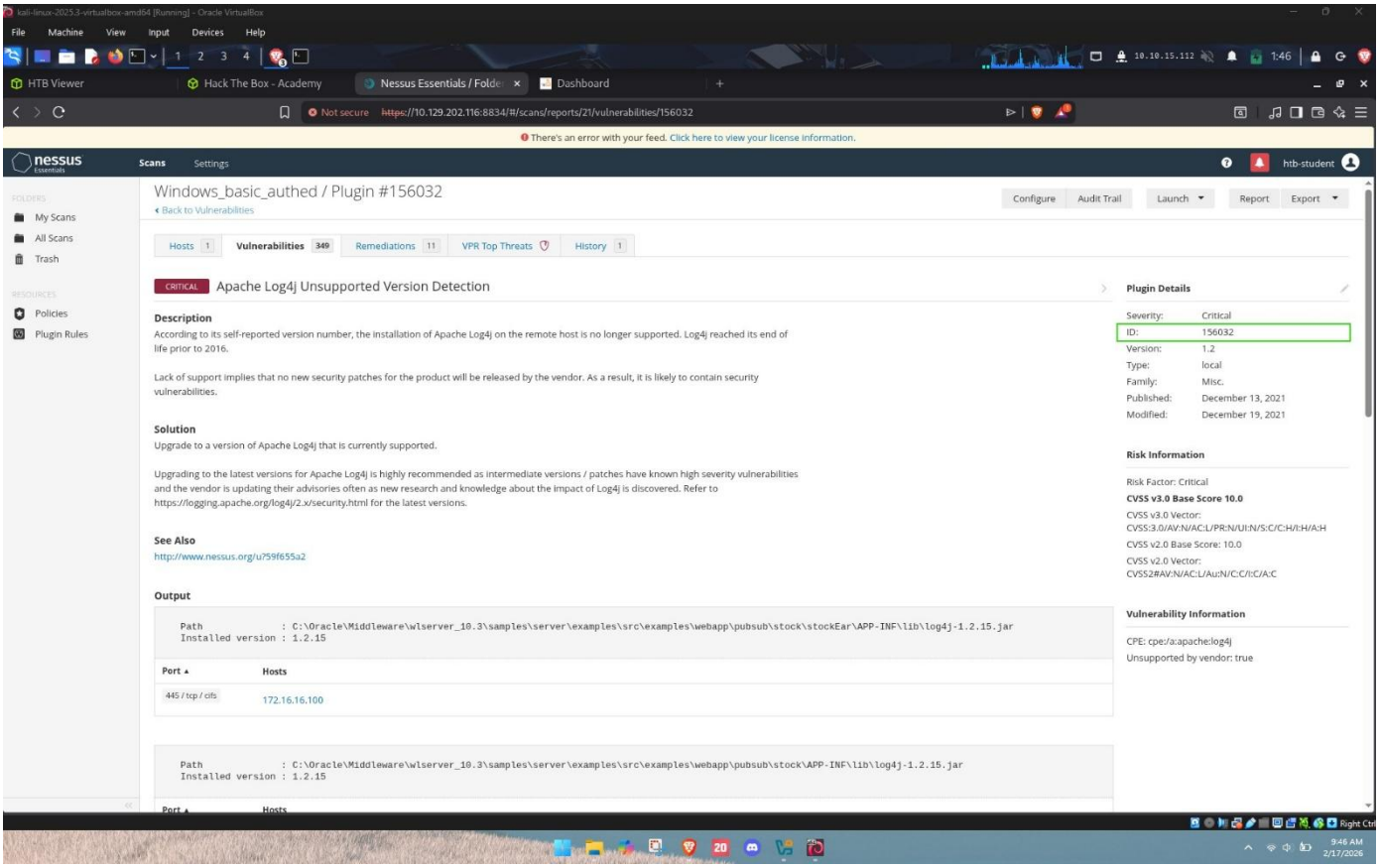


In order to find the answers to the questions, I had to use Nessus. The account that was used was already set up by hack the box academy; with credentials provided. So long as I was connected to the VPN, I was able to access Nessus through the IP provided (which in my case was 10.129.202.116 with the port number being 8834).

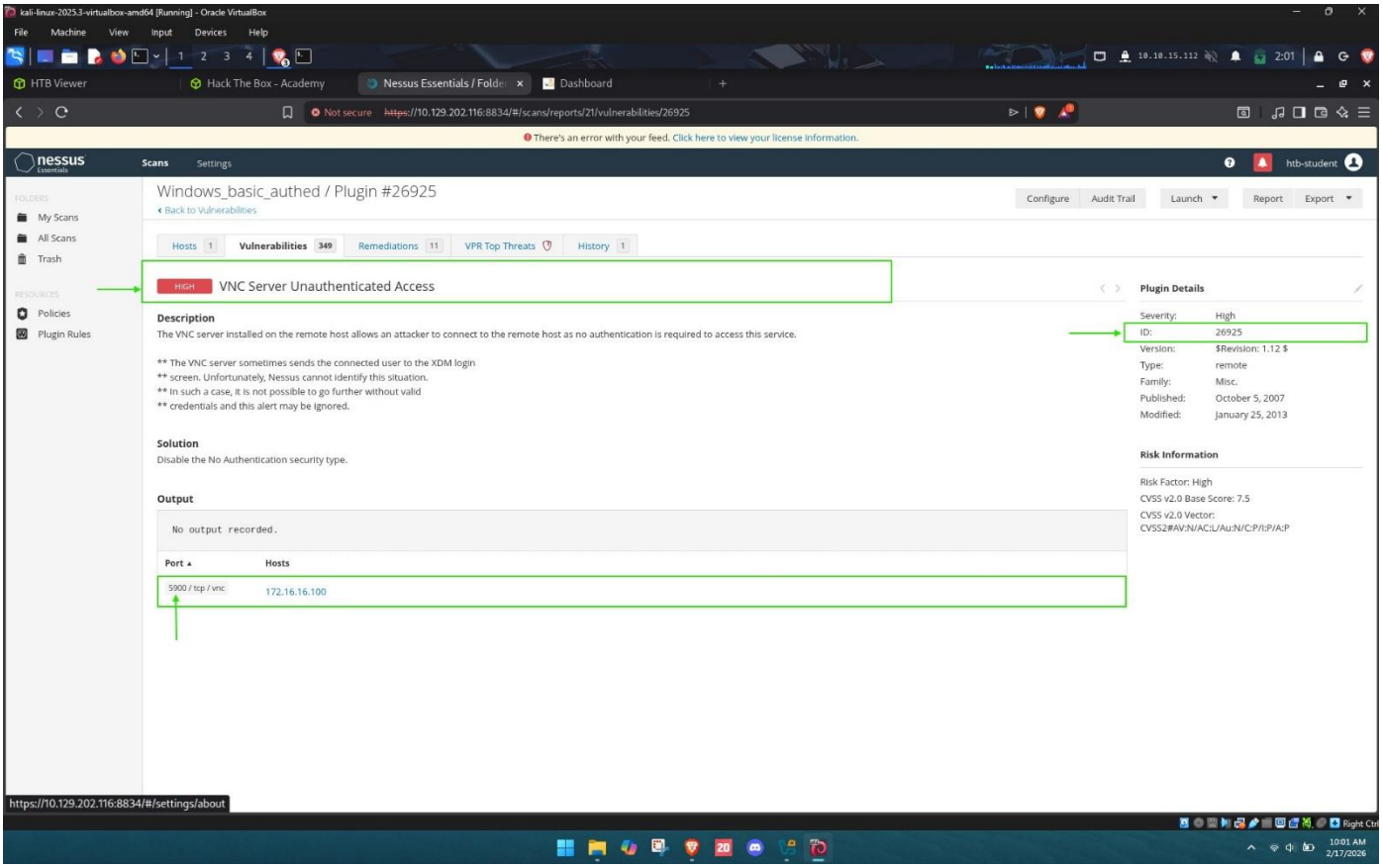
For the first question, I accessed the Windows scan and navigated to the vulnerabilities section. I then searched for SMB shares and was able to find the one as shown below. The second question's correct response was also seen in the same image marked by the arrow.



The third question prompted for the plugin ID for the highest criticality vulnerability in the scan, which I found by navigating to the vulnerability tab and selecting the first one; since they are arranged in order of severity. The answer is shown in the image below:



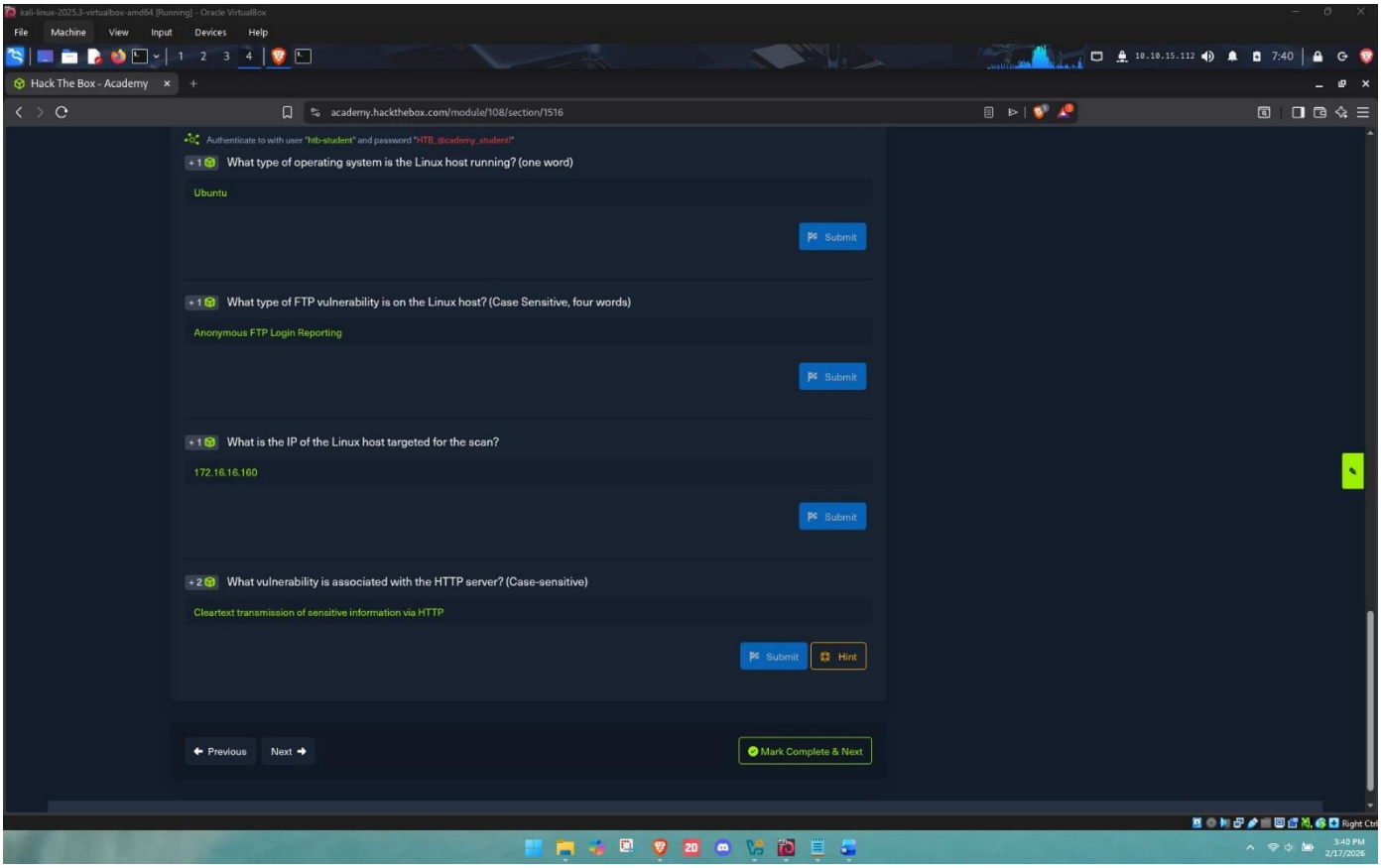
The next question was on the name of the vulnerability with the plugin ID 26925; which the answer is as shown below:



The last question was on the port the VNC server was running on in the scan, which is marked by the arrow in the image above.

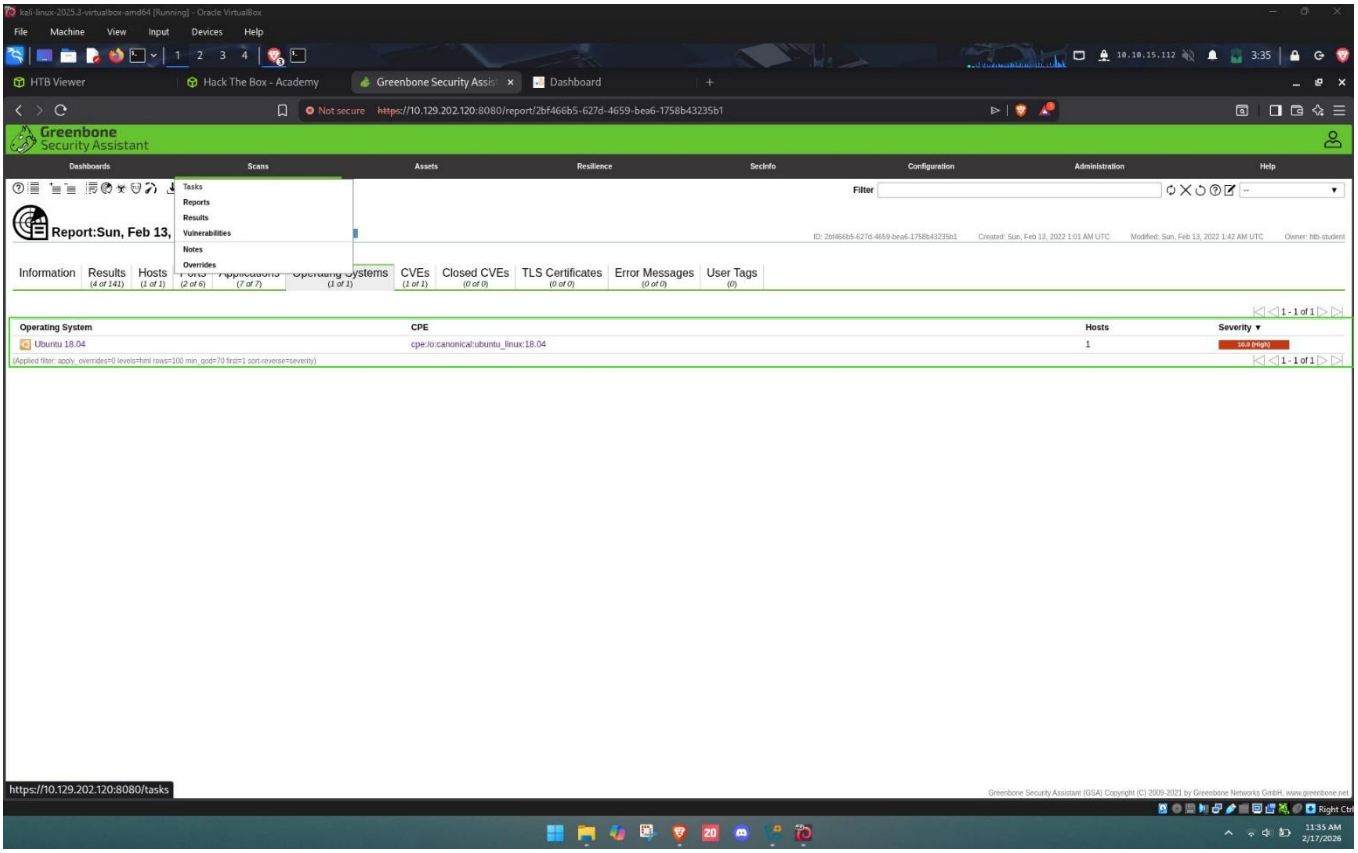
OpenVAS Skills Assessment

OpenVAS functions as an open-source security scanning tool which detects network and system and application security vulnerabilities. The system conducts automatic security scanning to find vulnerabilities and outdated software and configuration problems, which results in security reports that help organizations enhance their protection measures.

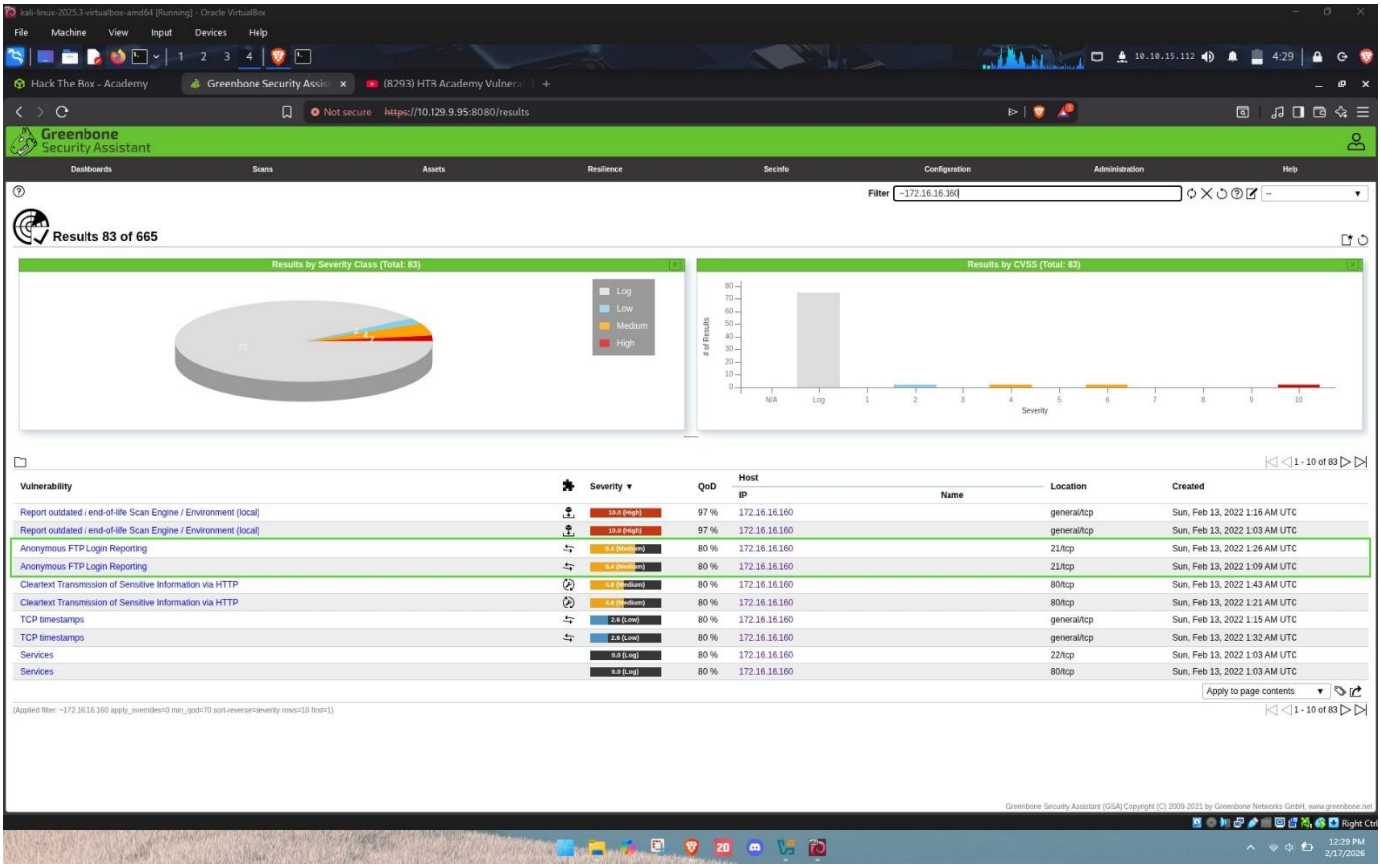


In order to answer these questions, I used the IP address provided (which was 10.129.202.120 through port number 8080) to access OpenVAS. I then used the credentials provided to login to the tool.

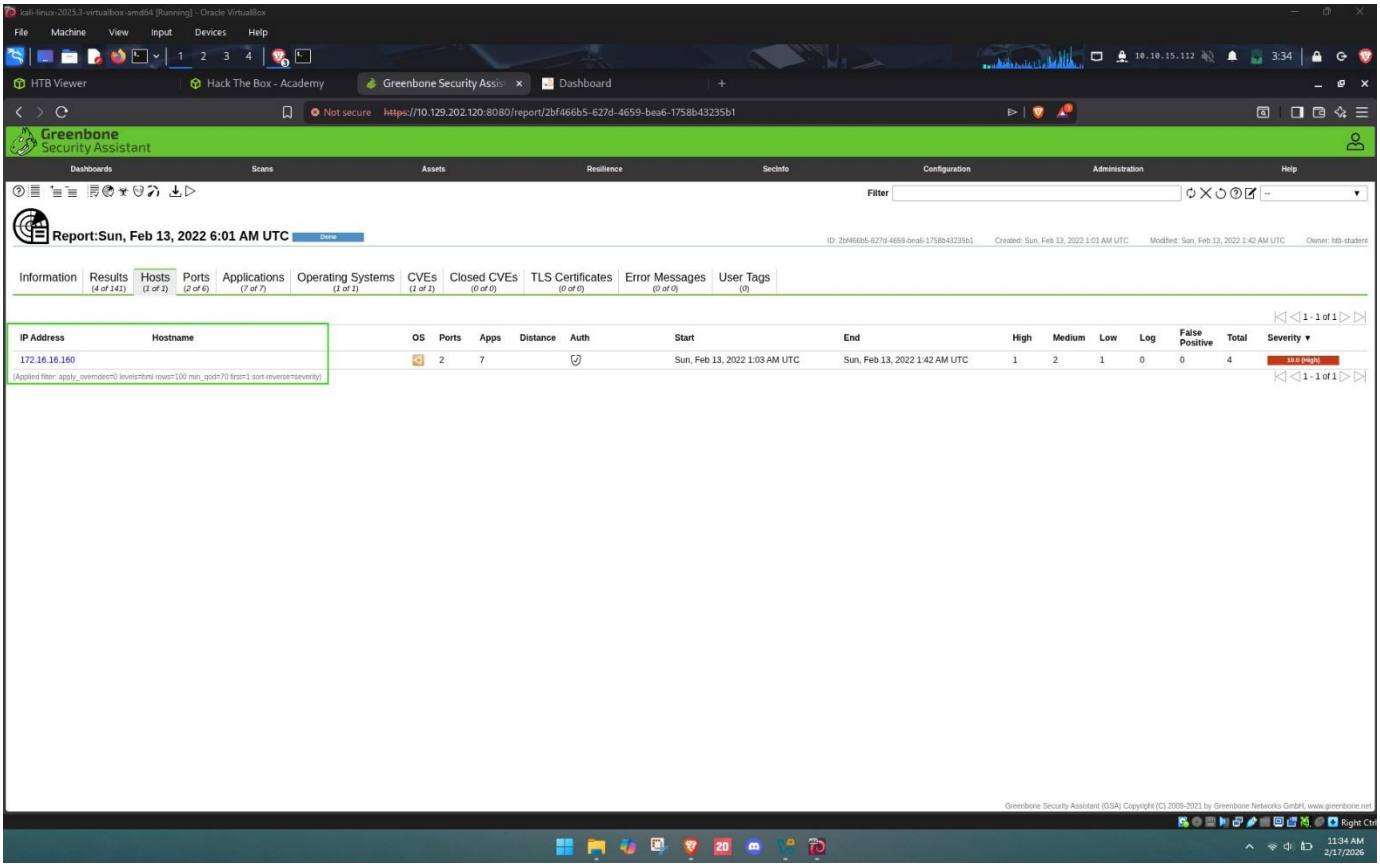
The first question was on the type of operating system that the Linux host was running. The result showed Ubuntu, as shown below:



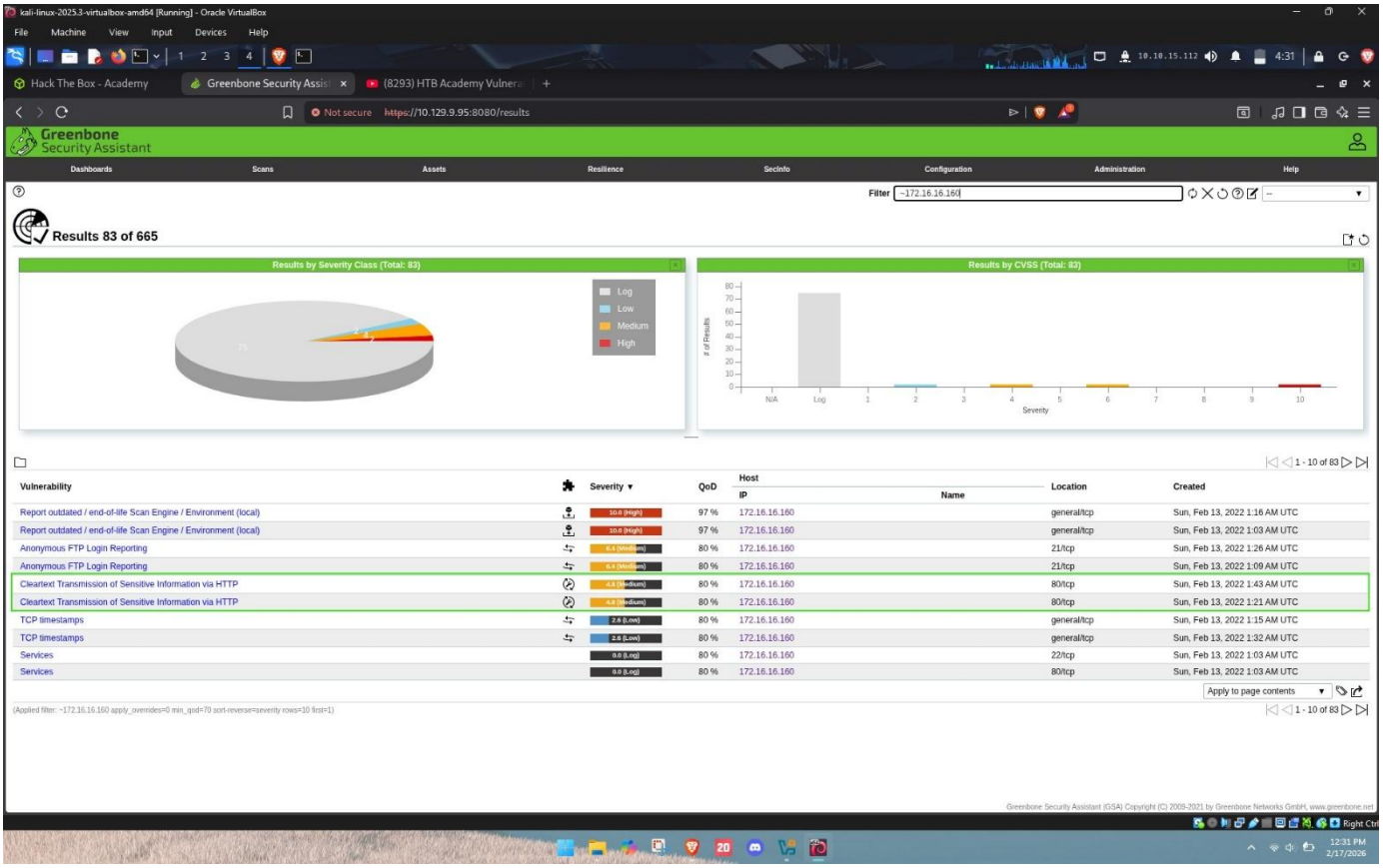
The second part was on the type of FTP vulnerability that is on the Linux host, which is as shown below. I typed FTP on the search bar and was able to see the result.



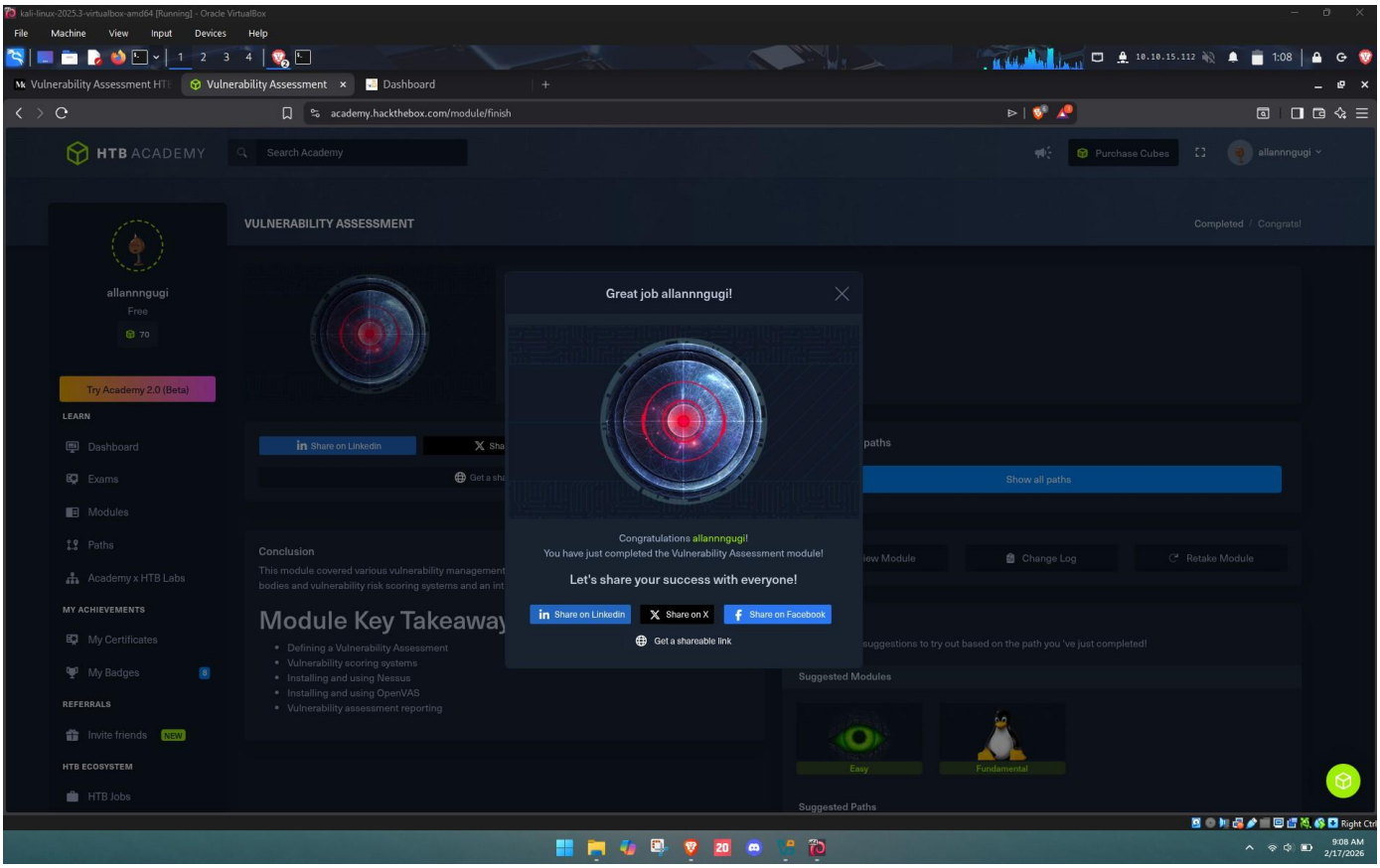
The third question was on the IP of the Linux host, which is shown below:



For the last question, I searched HTTP on the search bar, which gave me the result as shown below:


kali-linux-2025.3-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Hack The Box - Academy
Greenbone Security Assis
(8293) HTB Academy Vulnera
Not secure https://10.129.9.95:8080/results
Greenbone Security Assistant
Dashboards Scans Assets Resilience SecInfo Configuration Administration Help
Filter ~172.16.16.160
Results 83 of 665
Results by Severity Class (Total: 83)
Log
Low
Medium
High
Results by CVSS (Total: 83)
# of Results
Severity
1 - 10 of 83
Vulnerability Severity QoD Host IP Name Location Created
Report outdated / end-of-life Scan Engine / Environment (local) 10.0 (High) 97 % 172.16.16.160 general/tcp Sun, Feb 13, 2022 1:16 AM UTC
Report outdated / end-of-life Scan Engine / Environment (local) 10.0 (High) 97 % 172.16.16.160 general/tcp Sun, Feb 13, 2022 1:03 AM UTC
Anonymous FTP Login Reporting 6.4 (Medium) 80 % 172.16.16.160 21/tcp Sun, Feb 13, 2022 1:26 AM UTC
Anonymous FTP Login Reporting 6.4 (Medium) 80 % 172.16.16.160 21/tcp Sun, Feb 13, 2022 1:09 AM UTC
Cleartext Transmission of Sensitive Information via HTTP 4.8 (Medium) 80 % 172.16.16.160 80/tcp Sun, Feb 13, 2022 1:43 AM UTC
Cleartext Transmission of Sensitive Information via HTTP 4.8 (Medium) 80 % 172.16.16.160 80/tcp Sun, Feb 13, 2022 1:21 AM UTC
TCP timestamps 2.6 (Low) 80 % 172.16.16.160 general/tcp Sun, Feb 13, 2022 1:15 AM UTC
TCP timestamps 2.6 (Low) 80 % 172.16.16.160 general/tcp Sun, Feb 13, 2022 1:32 AM UTC
Services 0.0 (Log) 80 % 172.16.16.160 22/tcp Sun, Feb 13, 2022 1:03 AM UTC
Services 0.0 (Log) 80 % 172.16.16.160 80/tcp Sun, Feb 13, 2022 1:03 AM UTC
Apply to page contents
(Applied filter: ~172.16.16.160 apply_overrides=0 min_qod=70 sort-reverse=severity rows=10 first=1)
1 - 10 of 83
Greenbone Security Assistant (GSA) Copyright (C) 2009-2021 by Greenbone Networks GmbH, www.greenbone.net

Screenshot Showing Module Completion



Conclusion

Vulnerability Assessment in this module demonstrated how automated scanning tools can identify security weaknesses in systems and networks. The room required one to run vulnerability scans with correct settings in order to produce reliable scan results. The room displayed different methods for evaluating scan results which help organizations to identify their most critical security risks. The assessment demonstrated that organizations use vulnerability assessment to find and eliminate threats before they become actual security breaches.

Link Showing Module Completion

https://academy.hackthebox.com/achievement/2384006/108